

1 Basic convergence tests

Question 1. Match the series on the left with the appropriate convergence test and conclusion on the right, evaluating the series if asked.

Divergent by a variety of tests
 • p-series
 • comparison with $\frac{1}{n}$
 • Integral

$\lim_{n \rightarrow \infty} a_n \neq 0 \Rightarrow \sum_{n=1}^{\infty} a_n$ diverges
 example: $\sum_{n=1}^{\infty} \frac{1}{\sqrt{1+\frac{1}{n}}}$ diverges

$\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$ Diverges by the divergence test

$\sum_{n=1}^{\infty} \frac{1}{\sqrt{1+n}}$ It's a convergent geometric series that converges to 3

$\sum_{n=2}^{\infty} \frac{1}{n \ln n}$ It's a telescoping series that converges to 9/4

$\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ Converges by the p-series Test

$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ Diverges by the p-series Test

$\sum_{n=1}^{\infty} \frac{3^n}{4^n}$ Converges by the integral test

$\sum_{n=1}^{\infty} \frac{3}{n(n+2)}$ Diverges by the integral test

$\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+2} \right) = \frac{3}{2}$

$\sum_{n=1}^{\infty} \frac{1}{n^p} \rightarrow p > 1 \Rightarrow \text{converges}$
 $\rightarrow p \leq 1 \Rightarrow \text{diverges}$

$\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2} \leq \frac{1}{2(\ln 2)^2} + \int_2^{\infty} \frac{1}{x(\ln x)^2} dx$
 $= \int_{\ln 2}^{\infty} \frac{1}{y^2} dy = \frac{1}{\ln 2}$

$\sum_{n=2}^{\infty} \frac{1}{n(\ln n)} \geq \int_2^{\infty} \frac{1}{x \ln x} dx$
 $= \int_{\ln 2}^{\infty} \frac{1}{y} dy$ diverges

Question 2. The following boxed claim and proof are incorrect. Identify the mistake(s).

Claim: $\sum_{n=1}^{\infty} n^3 e^{-n} \leq 6$

Proof: The integral test tells us that

$\int_1^{\infty} x^3 e^{-x} dx \leq \sum_{n=1}^{\infty} n^3 e^{-n} \leq \int_0^{\infty} x^3 e^{-x} dx.$

We evaluate the right integral to find that

This is perfectly fine! $\rightarrow \int_0^{\infty} x^3 e^{-x} dx = [(-x^3 - 3x^2 - 6x - 6)e^{-x}]_0^{\infty} = 6.$

Hence, the inequality holds.

Integral test can only be applied when f is decreasing. Otherwise, there is no direct comparison between the series and integral.

$\sum_{n=1}^{\infty} f(n) \leq \int_0^{\infty} f(x) dx$ when f is decreasing.

no meaningful comparison when f is not decreasing.

Challenge: correctly prove that $\sum_{n=1}^{\infty} n^3 e^{-n}$ converges and find an actual upper bound. (Check your answer with the fact that $\sum_{n=1}^{\infty} n^3 e^{-n} \approx 6.0065$.)

2 Estimating errors

Let $a_n \geq 0$ be a nonnegative sequence. By estimating the infinite series $\sum_{n=1}^{\infty} a_n$ with a partial sum $\sum_{n=1}^N a_n$ to a precision of ϵ , we mean finding an integer N such that

$$\sum_{n=1}^{\infty} a_n - \sum_{n=1}^N a_n = \sum_{n=N+1}^{\infty} a_n < \epsilon.$$

This can be done by comparing the series to an integral. If f is a decreasing function such that $f(n) = a_n$, then integral test tells us that

$$\int_{N+1}^{\infty} f(x) dx \leq \sum_{n=N+1}^{\infty} a_n \leq \int_N^{\infty} f(x) dx,$$

so the task becomes finding an integer N such that

$$\sum_{n=N+1}^{\infty} a_n \leq \int_N^{\infty} f(x) dx < \epsilon.$$

Question 3 (Estimating errors). Estimate a value of N such that $\sum_{n=1}^N a_n$ approximates the infinite series $\sum_{n=1}^{\infty} a_n$ to within a precision of 0.03.

a) $a_n = \frac{1}{n^2}$ $\int_N^{\infty} \frac{1}{x^2} dx = \frac{1}{N} < 0.03 \Rightarrow N \text{ can be } 34$

b) $a_n = \frac{1}{(n+1)(\ln(n+1))^2}$

c) $a_n = \frac{1}{n(n+1)}$ (Challenge: use another method besides comparison with an integral)

$\int_N^{\infty} \frac{1}{x(x+1)} dx = \lim_{M \rightarrow \infty} \int_N^M \left(\frac{1}{x} - \frac{1}{x+1} \right) dx = \lim_{M \rightarrow \infty} (\ln M - \ln N - \ln(M+1) + \ln(N+1)) = \ln(N+1) - \ln N = \ln\left(1 + \frac{1}{N}\right) < 0.03$

3 Comparison tests

Question 4. The following boxed claim and proof are incorrect. Identify the mistake(s).

Claim: $\sum_{n=1}^{\infty} \frac{\sin(1/n)}{n}$ diverges.

Proof: Since $\sin(1/n) \leq 1$ for every positive integer n , we can apply direct comparison test with the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$. Since the harmonic series diverges, the original series must also diverge.

$\sin(\frac{1}{n}) \leq 1$ implies $\frac{\sin(1/n)}{n} \leq \frac{1}{n}$.
Growing slower than a divergent series doesn't necessarily imply that the series is divergent.

Challenge: show that $\sum_{n=1}^{\infty} \frac{\sin(1/n)}{n}$ converges. It grows at the same rate as $\frac{1}{n^2}$ because $\lim_{n \rightarrow \infty} \frac{\sin(1/n)}{1/n} = \lim_{n \rightarrow \infty} n \sin(1/n) = 1$. Since $\sum \frac{1}{n^2}$ is convergent, $\sum \frac{\sin(1/n)}{n}$ is convergent.

Question 5. Match the series with the appropriate conclusion and state the series used to compare.

$$\sum_{n=1}^{\infty} \frac{n}{3^n}$$

Converges because it grows at the same rate as the convergent series: $\sum \frac{1}{n^2}$ $\lim_{n \rightarrow \infty} \frac{n^2(n+2)}{n^2-2n+2} = 1$

$$\sum_{n=1}^{\infty} \frac{n+2}{n^3-2n+2}$$

Diverges because it grows at the same rate as the divergent series: $\sum \frac{1}{n}$ $\lim_{n \rightarrow \infty} n \sin(\frac{1}{n}) = 1$

$$\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right)$$

Converges because it grows slower than the convergent series: $\sum \frac{1}{2^n}$ $\lim_{n \rightarrow \infty} \frac{n \cdot 2^n}{3^n} = \lim_{n \rightarrow \infty} \frac{n}{(3/2)^n} = 0$

$$\sum_{n=1}^{\infty} \frac{1}{1 + \ln n}$$

Diverges because it grows faster than the divergent series: $\sum \frac{1}{n}$ $\lim_{n \rightarrow \infty} \frac{1 + \ln n}{n} = 0$